

Foundational MCQ - Answers

1. K-Nearest Neighbors (KNN)

1. What is the main idea behind KNN classification?

- A. Learn a sequence of decision rules
- B. Assign a class based on nearby training examples
- C. Fit a straight line to the data
- D. Build multiple decision trees

Answer: B. Assign a class based on nearby training examples

2. In KNN, what does K represent?

- A. The number of input features
- B. The number of classes
- C. The number of neighbors considered
- D. The number of training iterations

Answer: C. The number of neighbors considered

3. A KNN classifier has `n_neighbors=5`. How is a new sample usually classified?

- A. By the class of the first training sample
- B. By a vote among its five nearest training samples
- C. By averaging all training labels
- D. By selecting five random classes

Answer: B. By a vote among its five nearest training samples

4. Which Python class is used for KNN classification in scikit -learn?

- A. KMeans
- B. KNeighborsClassifier
- C. NearestCentroidRegressor
- D. NeighborTree

Answer: B. KNeighborsClassifier

5. Which import statement is appropriate for KNN classification?

- A. `from sklearn.neighbors import KNeighborsClassifier`
- B. `from sklearn.tree import KNeighborsClassifier`
- C. `from sklearn.linear_model import KNeighborsClassifier`
- D. `from sklearn.ensemble import KNeighborsClassifier`

Answer: A. `from sklearn.neighbors import KNeighborsClassifier`

6. Which method trains a KNN model in scikit -learn?

- A. `model.learn(X_train)`
- B. `model.fit(X_train, y_train)`
- C. `model.predict(X_train, y_train)`
- D. `model.train(y_train)`

Answer: B. `model.fit(X_train, y_train)`

7. Which method predicts labels for new samples?

- A. `model.fit(X_test)`

- B. `model.score(y_test)`
- C. `model.predict(X_test)`
- D. `model.transform(y_test)`

Answer: C. `model.predict(X_test)`

8. Why is feature scaling often important for KNN?

- A. KNN requires features to be integers
- B. Features with larger numerical ranges may dominate distance calculations
- C. Scaling increases the number of classes
- D. Scaling removes the need for training data

Answer: B. Features with larger numerical ranges may dominate distance calculations

9. Consider age ranging from 18 to 80 and annual income ranging from 10,000 to 2,000,000. What may happen if KNN uses these features without scaling?

- A. Age will automatically receive more weight
- B. Income may dominate the distance calculation
- C. KNN will ignore both features
- D. The model will become a decision tree

Answer: B. Income may dominate the distance calculation

10. Which tool is commonly used to standardize features before KNN?

- A. `StandardScaler`
- B. `LabelEncoder` on the target only
- C. `DecisionTreeClassifier`
- D. `confusion_matrix`

Answer: A. `StandardScaler`

11. What is a sensible way to combine scaling and KNN in Python?

- A. `Pipeline([("scaler", StandardScaler()), ("knn", KNeighborsClassifier())])`
- B. `KNeighborsClassifier(StandardScaler())`
- C. `StandardScaler(KNeighborsClassifier())`
- D. `Pipeline([("knn", KNeighborsClassifier()), ("scaler", StandardScaler())])`

Answer: A. `Pipeline([("scaler", StandardScaler()), ("knn", KNeighborsClassifier())])`

12. Why should a scaler be fitted only on the training data?

- A. To reduce the number of classes
- B. To prevent information from the test set entering model preparation
- C. To make KNN use Euclidean distance
- D. To remove all missing values

Answer: B. To prevent information from the test set entering model preparation

13. What generally happens when K is set to 1?

- A. The classifier considers the single closest training example
- B. The classifier uses all training examples
- C. The classifier becomes a regression model
- D. The classifier ignores distance

Answer: A. The classifier considers the single closest training example

14. A very small K can make KNN particularly sensitive to what?

- A. Feature names
- B. Noise and unusual training examples
- C. The order of Python imports
- D. The number of CPU cores only

Answer: B. Noise and unusual training examples

15. A very large K may cause which problem?

- A. Predictions may become overly smooth and miss local patterns
- B. The model memorizes every training label perfectly
- C. Distance calculations stop working
- D. The number of input features becomes zero

Answer: A. Predictions may become overly smooth and miss local patterns

16. What is the purpose of trying several K values with cross -validation?

- A. To choose a K that performs well on unseen data
- B. To change the target labels
- C. To avoid using a test set entirely
- D. To make all predictions identical

Answer: A. To choose a K that performs well on unseen data

17. For a binary classification problem, why might someone initially try an odd K?

- A. It guarantees perfect accuracy
- B. It can reduce voting ties
- C. It eliminates the need for scaling
- D. It makes KNN train faster in every case

Answer: B. It can reduce voting ties

18. What does weights="distance" do in KNeighborsClassifier?

- A. Gives closer neighbors greater influence
- B. Gives every neighbor zero influence
- C. Weights features according to their names
- D. Converts classification into regression

Answer: A. Gives closer neighbors greater influence

19. How does weights="uniform" treat the selected neighbors?

- A. All selected neighbors contribute equally to the vote
- B. Only the closest neighbor votes
- C. More distant neighbors always vote twice
- D. Each feature receives the same coefficient

Answer: A. All selected neighbors contribute equally to the vote

20. Which distance measure is commonly associated with straight -line distance between points?

- A. Euclidean distance
- B. Gini impurity
- C. Log loss
- D. Information gain

Answer: A. Euclidean distance

21. Which distance measure sums absolute differences across features?

- A. Manhattan distance
- B. Euclidean distance
- C. Entropy
- D. Precision

Answer: A. Manhattan distance

22. In scikit -learn KNN, what does metric="manhattan" change?

- A. The distance used to find neighbors
- B. The number of target classes
- C. The training labels
- D. The evaluation metric returned by score

Answer: A. The distance used to find neighbors

23. What happens when n_neighbors is larger than the number of available training samples?

- A. Prediction can fail because there are too few samples
- B. KNN creates new training samples automatically
- C. The model switches to Naive Bayes
- D. KNN silently removes all features

Answer: A. Prediction can fail because there are too few samples

24. Which type of task is KNeighborsRegressor designed for?

- A. Predicting a continuous value
- B. Grouping data without labels
- C. Predicting only text labels
- D. Selecting important features

Answer: A. Predicting a continuous value

25. How does basic KNN regression predict a value?

- A. It combines target values of nearby training samples
- B. It chooses the most common class label
- C. It fits one global linear equation
- D. It counts the number of decision tree leaves

Answer: A. It combines target values of nearby training samples

26. Which is an appropriate KNN regression problem?

- A. Predicting house price from features of similar houses
- B. Assigning an email to spam or not spam
- C. Identifying whether an image contains a cat
- D. Classifying a tumor as benign or malignant

Answer: A. Predicting house price from features of similar houses

27. Which is an appropriate KNN classification problem?

- A. Predicting a patient's exact blood pressure
- B. Predicting whether a patient belongs to a risk category
- C. Predicting a product's selling price

D. Predicting tomorrow's temperature

Answer: A. Predicting a patient's exact blood pressure

28. Why can KNN prediction become slow on a very large training set?

- A. It may need to search many stored training samples for neighbors
- B. It retrains a neural network for each sample
- C. It builds a new random forest for each prediction
- D. It must calculate entropy for every feature

Answer: A. It may need to search many stored training samples for neighbors

29. What does it mean that KNN is an instance -based method?

- A. It uses stored training examples when making predictions
- B. It creates one rule for every class and discards the data
- C. It requires exactly one feature
- D. It trains only on images

Answer: A. It uses stored training examples when making predictions

30. How can irrelevant features affect KNN?

- A. They can distort which samples appear close
- B. They are always ignored automatically
- C. They increase the number of labels
- D. They turn continuous targets into categories

Answer: A. They can distort which samples appear close

31. What is the "curse of dimensionality" in the context of KNN?

- A. Distances may become less informative as the number of features grows
- B. Python cannot store more than three features
- C. KNN requires one neighbor per feature
- D. High -dimensional data always gives perfect predictions

Answer: A. Distances may become less informative as the number of features grows

32. Two records have identical feature values but different class labels. What difficulty can this create for KNN?

- A. The nearby evidence conflicts
- B. KNN cannot store either record
- C. The number of features doubles
- D. Distance becomes negative

Answer: A. The nearby evidence conflicts

33. Which scikit -learn call returns class probabilities from a fitted KNN classifier?

- A. `model.predict_proba(X_test)`
- B. `model.predict_probability(y_test)`
- C. `model.fit_proba(X_test)`
- D. `model.class_probabilities()`

Answer: A. `model.predict_proba(X_test)`

34. In a three -neighbor uniform -voting KNN model, two neighbors belong to class A and one to class

- A. A
- B. B

- C. Both classes are always returned
- D. The prediction cannot be made

Answer: A. A

35. What is a reasonable first step if one feature is measured in millimeters and another in kilometers?

- A. Scale the features appropriately
- B. Increase K to the number of training samples
- C. Remove the target column
- D. Use training accuracy as the only evaluation

Answer: A. Scale the features appropriately

36. What does `train_test_split(X, y, test_size=0.2)` commonly do?

- A. Reserves approximately 20% of samples for testing
- B. Uses 20 features for testing
- C. Removes 20% of the classes
- D. Sets K to 0.2

Answer: A. Reserves approximately 20% of samples for testing

37. Why should test samples remain separate while selecting K?

- A. To keep the final test evaluation independent
- B. To make the training set smaller for no reason
- C. To force Euclidean distance
- D. To prevent the use of cross-validation

Answer: A. To keep the final test evaluation independent

38. Which metric is useful when classifying a rare disease and missing a positive case is especially costly?

- A. Recall for the positive class
- B. Training set size alone
- C. Number of features alone
- D. Prediction time alone

Answer: A. Recall for the positive class

39. What does `model.score(X_test, y_test)` return for `KNeighborsClassifier` by default?

- A. Classification accuracy
- B. Mean squared error
- C. Number of neighbors
- D. Euclidean distance

Answer: A. Classification accuracy

40. Why can accuracy be misleading on a highly imbalanced classification dataset?

- A. Predicting the majority class can produce high accuracy while missing the minority class
- B. Accuracy always becomes negative
- C. KNN cannot use imbalanced data
- D. Accuracy is defined only for regression

Answer: A. Predicting the majority class can produce high accuracy while missing the minority class

41. A dataset contains a categorical feature such as blood type. What should you consider before applying standard distance -based KNN?

- A. How to encode the category and whether the resulting distance is meaningful
- B. Converting the target to a floating -point value
- C. Setting every blood type to the same number
- D. Increasing the number of test samples

Answer: A. How to encode the category and whether the resulting distance is meaningful

42. Why is assigning arbitrary numbers such as 1, 2, and 3 to unordered categories potentially problematic for KNN?

- A. The numbers imply distances and ordering that may not exist
- B. KNN accepts only strings
- C. Numeric features are ignored
- D. The target must contain three classes

Answer: A. The numbers imply distances and ordering that may not exist

43. What is the likely effect of adding many unrelated random features to a KNN model?

- A. Neighbor selection may become less reliable
- B. The model must always improve
- C. K automatically becomes 1
- D. Scaling becomes impossible

Answer: A. Neighbor selection may become less reliable

44. What does GridSearchCV help you do with a KNN pipeline?

- A. Evaluate candidate hyperparameters using cross -validation
- B. Create target labels from scratch
- C. Replace the test set with predictions
- D. Draw a decision tree

Answer: A. Evaluate candidate hyperparameters using cross -validation

45. Which parameter would you vary to study how many neighbors influence predictions?

- A. n_neighbors
- B. random_state
- C. max_depth
- D. C

Answer: A. n_neighbors

46. What is a limitation of applying ordinary KNN directly to raw, high -resolution images?

- A. Pixel -space distance may not match meaningful visual similarity
- B. Images cannot be converted into arrays
- C. KNN accepts only two input features
- D. KNN always returns continuous values

Answer: A. Pixel -space distance may not match meaningful visual similarity

47. If closer neighbors should matter more, which setting is most relevant?

- A. weights="distance"
- B. weights="uniform"

- C. max_depth=3
- D. criterion="entropy"

Answer: A. weights="distance"

48. When is KNN especially intuitive as a baseline?

- A. When similar feature profiles are expected to have similar outcomes
- B. When no labeled training examples are available
- C. When every prediction must be a linear equation
- D. When the target has no relationship with the features

Answer: A. When similar feature profiles are expected to have similar outcomes

49. In model.fit(X_train, y_train), what does X_train contain?

- A. Training input features
- B. Predicted probabilities
- C. Test labels
- D. Evaluation scores

Answer: A. Training input features

50. In model.fit(X_train, y_train), what does y_train contain?

- A. Training target values or class labels
- B. Feature scaling parameters only
- C. Test predictions
- D. Distance metrics

Answer: A. Training target values or class labels

2. Naive Bayes

1. What is Naive Bayes primarily used for?

- A. Supervised classification
- B. Drawing decision trees
- C. Clustering without labels
- D. Sorting a dataset by date

Answer: A. Supervised classification

2. What does the "naive" assumption refer to?

- A. Features are treated as conditionally independent given the class
- B. The model assumes every class is equally likely in all datasets
- C. The model uses no training data
- D. The model always predicts the first class

Answer: A. Features are treated as conditionally independent given the class

3. Which theorem provides the foundation for Naive Bayes?

- A. Bayes' theorem
- B. Pythagoras' theorem
- C. The central limit theorem only
- D. The law of large numbers only

Answer: A. Bayes' theorem

4. In a spam classifier, what does the model estimate?

- A. The probability of a class given features of an email
- B. Only the number of characters in an email
- C. The Euclidean distance between all emails
- D. The depth of a decision tree

Answer: A. The probability of a class given features of an email

5. Which Python class is commonly used for Naive Bayes with continuous features assumed to follow class-specific Gaussian distributions?

- A. GaussianNB
- B. MultinomialNB
- C. BernoulliNB
- D. KNeighborsClassifier

Answer: A. GaussianNB

6. Which Naive Bayes variant is commonly used with word -count features in text classification?

- A. MultinomialNB
- B. GaussianNB
- C. BernoulliNB only for every text dataset
- D. DecisionTreeRegressor

Answer: A. MultinomialNB

7. Which variant naturally models binary features such as whether a word appears?

- A. BernoulliNB
- B. GaussianNB
- C. LinearRegression
- D. RandomForestRegressor

Answer: A. BernoulliNB

8. Which import statement is appropriate for Gaussian Naive Bayes?

- A. from sklearn.naive_bayes import GaussianNB
- B. from sklearn.neighbors import GaussianNB
- C. from sklearn.tree import GaussianNB
- D. from sklearn.metrics import GaussianNB

Answer: A. from sklearn.naive_bayes import GaussianNB

9. Which statement creates a Gaussian Naive Bayes model?

- A. model = GaussianNB()
- B. model = GaussianNB.fit()
- C. model = BayesGaussianClassifier
- D. model = GaussianNB.predict()

Answer: A. model = GaussianNB()

10. What does model.fit(X_train, y_train) do?

- A. Learns parameters from labeled training data
- B. Predicts labels for the test set
- C. Calculates only test accuracy

D. Removes duplicate records automatically

Answer: A. Learns parameters from labeled training data

11. What does `model.predict(X_test)` return?

- A. Predicted class labels
- B. A confusion matrix
- C. The training dataset
- D. A list of feature names

Answer: A. Predicted class labels

12. What does `model.predict_proba(X_test)` return?

- A. Estimated probabilities for each class
- B. Only distances between samples
- C. A new training set
- D. A decision tree diagram

Answer: A. Estimated probabilities for each class

13. In Bayes' theorem, what is a prior probability?

- A. The probability of a class before considering the current sample's features
- B. The probability that a feature is missing
- C. The final prediction for a sample
- D. The test accuracy

Answer: A. The probability of a class before considering the current sample's features

14. In a dataset where 90% of emails are legitimate, what can the model learn about class priors?

- A. Legitimate email is more common in the training data
- B. Both classes must have a prior of 0.5
- C. Spam is always more probable
- D. Priors are unrelated to class frequency

Answer: A. Legitimate email is more common in the training data

15. What is a likelihood in Naive Bayes?

- A. How probable the observed features are under a particular class
- B. The number of model parameters
- C. The distance to the nearest sample
- D. The final accuracy on a test set

Answer: A. How probable the observed features are under a particular class

16. Which application is a common use of Naive Bayes?

- A. Document topic classification
- B. Image resizing
- C. Database backup
- D. Sorting files alphabetically

Answer: A. Document topic classification

17. Why is Naive Bayes often a useful baseline for text classification?

- A. It can work efficiently with high -dimensional word features
- B. It requires no labels

- C. It always understands word meaning and context perfectly
- D. It needs every document to have the same number of words

Answer: A. It can work efficiently with high -dimensional word features

18. What does CountVectorizer generally produce from text documents?

- A. Numeric features representing token counts
- B. A decision tree
- C. A list of class predictions
- D. A set of images

Answer: A. Numeric features representing token counts

19. Which pipeline is suitable for a basic word -count spam classifier?

- A. CountVectorizer() followed by MultinomialNB()
- B. GaussianNB() followed by CountVectorizer()
- C. KNeighborsClassifier() followed by CountVectorizer()
- D. confusion_matrix() followed by MultinomialNB()

Answer: A. CountVectorizer() followed by MultinomialNB()

20. Why must text be converted into numeric features before fitting a standard scikit -learn Naive Bayes classifier?

- A. The classifier expects a numerical feature representation
- B. The classifier can only predict numbers
- C. Text labels are forbidden
- D. All words must be removed

Answer: A. The classifier expects a numerical feature representation

21. What is Laplace smoothing intended to address?

- A. Zero probabilities caused by unseen feature -class combinations
- B. Incorrect train -test splitting
- C. Missing target labels in all datasets
- D. Excessively deep decision trees

Answer: A. Zero probabilities caused by unseen feature -class combinations

22. In MultinomialNB(alpha=1.0), what does alpha control?

- A. Additive smoothing
- B. Number of nearest neighbors
- C. Maximum tree depth
- D. Regularization strength C

Answer: A. Additive smoothing

23. What may happen without smoothing when a word has never appeared in training examples of a class?

- A. The estimated likelihood for that word under the class may be zero
- B. The word is automatically assigned to every class
- C. All training labels are deleted
- D. The model becomes a regression model

Answer: A. The estimated likelihood for that word under the class may be zero

24. Which statement about the independence assumption is most accurate?

- A. It is an approximation that may still yield useful predictions

- B. It must be exactly true for the model to make any prediction
- C. It means features are independent of the target
- D. It means training samples cannot share features

Answer: A. It is an approximation that may still yield useful predictions

25. Consider the words "high" and "fever" in medical notes. What might violate the naive assumption?

- A. Their occurrences may be related even after accounting for the class
- B. Both words contain letters
- C. One word is shorter than the other
- D. Both words can be converted into numbers

Answer: A. Their occurrences may be related even after accounting for the class

26. Which dataset is most naturally suited to GaussianNB among these choices?

- A. Continuous measurements such as temperature and blood pressure
- B. Raw text strings without numerical conversion
- C. Document word counts with thousands of tokens
- D. Binary word -presence features only

Answer: A. Continuous measurements such as temperature and blood pressure

27. Does "Gaussian" in GaussianNB describe the assumed distribution of each continuous feature within each class?

- A. Yes
- B. No, it specifies the number of classes
- C. No, it specifies the train -test split
- D. No, it specifies the evaluation metric

Answer: A. Yes

28. A model predicts "positive" with probability 0.8 and "negative" with probability 0.2. What is its usual predicted class?

- A. Positive
- B. Negative
- C. Both classes
- D. Neither class

Answer: A. Positive

29. What does `model.classes_` show after fitting a scikit -learn classifier?

- A. The class labels and their probability -column order
- B. The number of input features only
- C. The feature importance scores
- D. The original training samples

Answer: A. The class labels and their probability -column order

30. Why should a medical screening classifier be evaluated with more than accuracy?

- A. Missing positive cases and issuing false alarms can have different consequences
- B. Accuracy cannot be computed for Naive Bayes
- C. Every dataset must have exactly three classes
- D. Naive Bayes always predicts one class

Answer: A. Missing positive cases and issuing false alarms can have different consequences

31. Which metric measures the fraction of actual positive cases identified?

- A. Recall
- B. Specificity
- C. Mean squared error
- D. Tree depth

Answer: A. Recall

32. Which metric measures the fraction of predicted positive cases that are truly positive?

- A. Precision
- B. Recall
- C. Number of features
- D. Training time

Answer: A. Precision

33. What can a confusion matrix show for a binary Naive Bayes classifier?

- A. Counts of correct and incorrect predictions by class
- B. The distribution assumed for each feature
- C. The exact words in every document
- D. The source code of the model

Answer: A. Counts of correct and incorrect predictions by class

34. Why should CountVectorizer be fitted on training documents rather than all documents before splitting?

- A. To avoid learning vocabulary information from the test set
- B. To make the test set larger
- C. To increase the number of classes
- D. To guarantee 100% accuracy

Answer: A. To avoid learning vocabulary information from the test set

35. What is a practical benefit of using a scikit -learn Pipeline for text classification?

- A. It keeps text transformation and classification together during training and evaluation
- B. It removes the need for a target variable
- C. It ensures all predictions are correct
- D. It converts classification into clustering

Answer: A. It keeps text transformation and classification together during training and evaluation

36. If a word occurs frequently in both spam and legitimate emails, what is likely true?

- A. That word alone may provide limited discrimination between the classes
- B. The model must always classify the email as spam
- C. The word will be deleted automatically
- D. The training process must fail

Answer: A. That word alone may provide limited discrimination between the classes

37. If a word is common in spam but rare in legitimate emails, how might it affect prediction?

- A. Its presence can support the spam class
- B. Its presence always makes both classes equally probable
- C. It prevents any class prediction
- D. It reduces the number of input documents

Answer: A. Its presence can support the spam class

38. What is the distinction between MultinomialNB and BernoulliNB in a typical text application?

- A. The former can use token counts; the latter commonly uses binary presence indicators
- B. The former is for regression; the latter is for clustering
- C. The former needs images; the latter needs audio
- D. They are both decision tree algorithms

Answer: A. The former can use token counts; the latter commonly uses binary presence indicators

39. Which target is suitable for Naive Bayes classification?

- A. "Spam" versus "not spam"
- B. Exact house price in rupees
- C. Exact patient height in centimeters
- D. Continuous rainfall amount

Answer: A. "Spam" versus "not spam"

40. What should you do if the training and deployment data have substantially different class proportions?

- A. Examine whether changed priors affect performance and probability estimates
- B. Assume performance must be identical
- C. Remove all class labels
- D. Increase tree depth

Answer: A. Examine whether changed priors affect performance and probability estimates

41. What does an estimated probability of 0.9 from predict_proba require before being trusted as a real-world risk estimate?

- A. Calibration and evaluation on representative data
- B. Only a larger Python font
- C. Changing every feature to a string
- D. Setting the test size to zero

Answer: A. Calibration and evaluation on representative data

42. Can Naive Bayes handle a classification problem with more than two classes?

- A. Yes
- B. No, it is always binary
- C. Only if K is odd
- D. Only with a random forest

Answer: A. Yes

43. Which statement best describes the relationship between features and class in Naive Bayes?

- A. The model uses feature evidence to compare candidate classes
- B. The model ignores all feature values
- C. The model predicts the class from row order
- D. The model requires one feature for each class

Answer: A. The model uses feature evidence to compare candidate classes

44. What does GaussianNB estimate from the training data for each feature and class?

- A. Parameters such as mean and variance
- B. Only tree depth

- C. Nearest -neighbor indices
- D. A sequence of gradient updates

Answer: A. Parameters such as mean and variance

45. What is a limitation when strongly related features are included in Naive Bayes?

- A. Their evidence can effectively be counted more than intended
- B. They are always merged automatically
- C. The target labels are erased
- D. The classifier cannot output probabilities

Answer: A. Their evidence can effectively be counted more than intended

46. Why is a held -out test set useful?

- A. It estimates how the trained model performs on unseen examples
- B. It is used to calculate training priors
- C. It replaces feature extraction
- D. It guarantees deployment performance

Answer: A. It estimates how the trained model performs on unseen examples

47. What does `classification_report(y_test, y_pred)` commonly display?

- A. Precision, recall, F1 -score, and support
- B. Only a histogram of feature values
- C. The source code of the model
- D. The optimal learning rate

Answer: A. Precision, recall, F1 -score, and support

48. A model marks every email as "not spam" in a dataset with very little spam. What should be checked carefully?

- A. Performance on the spam class
- B. Only overall accuracy
- C. Maximum tree depth
- D. Euclidean distance

Answer: A. Performance on the spam class

49. Which feature representation would be appropriate for using MultinomialNB on documents?

- A. Nonnegative token -count features
- B. Raw Python strings passed directly as numeric arrays
- C. Arbitrary negative word counts
- D. Image file paths only

Answer: A. Nonnegative token -count features

50. When would Naive Bayes be a reasonable first model?

- A. When building a fast, interpretable baseline for labeled text classification
- B. When no input features exist
- C. When the output must be an image
- D. When every feature relationship must be modeled exactly

Answer: A. When building a fast, interpretable baseline for labeled text classification

3. Decision Trees

1. How does a decision tree classifier make a prediction?

- A. It follows feature -based splits from the root to a leaf
- B. It averages distances to all samples
- C. It multiplies word likelihoods only
- D. It always predicts the most recent label

Answer: A. It follows feature -based splits from the root to a leaf

2. What does an internal node of a decision tree usually represent?

- A. A test or split on a feature
- B. The final prediction only
- C. A missing dataset
- D. A test accuracy value

Answer: A. A test or split on a feature

3. What does a leaf node usually contain?

- A. A prediction or class distribution
- B. Another training dataset
- C. A Python import statement
- D. A distance metric

Answer: A. A prediction or class distribution

4. Which scikit -learn class builds a decision tree for classification?

- A. DecisionTreeClassifier
- B. DecisionTreeRegressor
- C. KNeighborsClassifier
- D. LogisticRegression

Answer: A. DecisionTreeClassifier

5. Which scikit -learn class builds a decision tree for continuous -value prediction?

- A. DecisionTreeRegressor
- B. DecisionTreeClassifier
- C. GaussianNB
- D. RandomForestClassifier

Answer: A. DecisionTreeRegressor

6. Which import statement is appropriate for tree classification?

- A. from sklearn.tree import DecisionTreeClassifier
- B. from sklearn.ensemble import DecisionTreeClassifier
- C. from sklearn.neighbors import DecisionTreeClassifier
- D. from sklearn.metrics import DecisionTreeClassifier

Answer: A. from sklearn.tree import DecisionTreeClassifier

7. What does model.fit(X_train, y_train) do for a decision tree?

- A. Learns splits from labeled training data
- B. Draws the tree without learning

- C. Predicts labels for test data only
- D. Calculates the confusion matrix

Answer: A. Learns splits from labeled training data

8. What does `model.predict(X_test)` return for a decision tree classifier?

- A. Predicted class labels
- B. Only tree depth
- C. Training samples
- D. Feature names only

Answer: A. Predicted class labels

9. What is the root node?

- A. The first split at the top of the tree
- B. The final leaf on the left
- C. The test dataset
- D. The most frequent target value

Answer: A. The first split at the top of the tree

10. What is the purpose of selecting a split during tree training?

- A. To separate examples into groups that are more useful for prediction
- B. To randomly rename features
- C. To standardize every feature
- D. To remove the target labels

Answer: A. To separate examples into groups that are more useful for prediction

11. In classification trees, what does Gini impurity measure?

- A. How mixed the class labels are within a node
- B. The physical distance between records
- C. The number of missing features
- D. The speed of Python execution

Answer: A. How mixed the class labels are within a node

12. A node contains examples from only one class. How would you describe it?

- A. Pure
- B. Untrained
- C. Standardized
- D. Randomized

Answer: A. Pure

13. What does `criterion="entropy"` change in `DecisionTreeClassifier`?

- A. The measure used to evaluate candidate classification splits
- B. The number of training rows
- C. The target variable type
- D. The Python package being imported

Answer: A. The measure used to evaluate candidate classification splits

14. What does `max_depth=3` limit?

- A. How deep the tree can grow

- B. The number of input features to exactly three
- C. The number of training samples to three
- D. The number of classes to three

Answer: A. How deep the tree can grow

15. Why might a very deep decision tree overfit?

- A. It can learn narrow patterns specific to the training samples
- B. It cannot split the data
- C. It is forced to predict one class
- D. It always uses too few features

Answer: A. It can learn narrow patterns specific to the training samples

16. What is a likely symptom of overfitting?

- A. Very high training performance but much lower test performance
- B. Identical performance on all datasets by definition
- C. No training predictions
- D. No leaf nodes

Answer: A. Very high training performance but much lower test performance

17. What does min_samples_leaf control?

- A. The minimum number of training samples required in a leaf
- B. The maximum number of trees
- C. The minimum number of features in the dataset
- D. The test -set fraction

Answer: A. The minimum number of training samples required in a leaf

18. How can increasing min_samples_leaf affect a tree?

- A. It can prevent very small, specialized leaves
- B. It always makes the tree deeper
- C. It turns the tree into KNN
- D. It changes every continuous feature into a category

Answer: A. It can prevent very small, specialized leaves

19. What does min_samples_split control?

- A. The minimum number of samples required for a node to be considered for splitting
- B. The number of test sets
- C. The number of class labels
- D. The number of trees in an ensemble

Answer: A. The minimum number of samples required for a node to be considered for splitting

20. Why is feature scaling usually less important for a standard decision tree than for KNN?

- A. Tree splits depend on feature thresholds rather than distances between whole samples
- B. Trees accept only standardized inputs
- C. Trees ignore numerical features
- D. Scaling always changes every tree prediction

Answer: A. Tree splits depend on feature thresholds rather than distances between whole samples

21. A tree uses the rule $\text{age} \leq 45$. What happens to a sample with age 40?

- A. It follows the branch for values satisfying the condition
- B. It is removed from the dataset
- C. It follows both branches equally
- D. It cannot be classified

Answer: A. It follows the branch for values satisfying the condition

22. What is a practical strength of a small decision tree?

- A. Its decision rules can often be inspected
- B. It always has the highest possible accuracy
- C. It needs no training data
- D. It automatically explains causation

Answer: A. Its decision rules can often be inspected

23. Which function can visualize a fitted scikit -learn decision tree?

- A. `plot_tree`
- B. `train_test_split`
- C. `accuracy_score`
- D. `StandardScaler`

Answer: A. `plot_tree`

24. What does `feature_importances_` provide for a fitted tree?

- A. Impurity -based importance scores for input features
- B. Proof that a feature causes the outcome
- C. The original feature values
- D. Class probabilities for each test sample

Answer: A. Impurity -based importance scores for input features

25. Why should feature importance be interpreted cautiously?

- A. Importance in the fitted model does not establish a causal relationship
- B. It always sums to a negative number
- C. It is available only before fitting
- D. It reports the test accuracy of each class

Answer: A. Importance in the fitted model does not establish a causal relationship

26. Which is an appropriate decision tree classification task?

- A. Predicting whether a loan application belongs to an approval category
- B. Predicting the exact loan amount as a class label without defining categories
- C. Counting the number of files in a folder
- D. Displaying a spreadsheet

Answer: A. Predicting whether a loan application belongs to an approval category

27. Which is an appropriate decision tree regression task?

- A. Predicting a house's sale price
- B. Classifying an email as spam
- C. Identifying a disease category
- D. Predicting whether a transaction is fraudulent

Answer: A. Predicting a house's sale price

28. What does a classification tree commonly predict at a leaf?

- A. A class based on training examples reaching that leaf
- B. The Euclidean distance to the root
- C. The number of Python packages installed
- D. The maximum input value in the entire dataset

Answer: A. A class based on training examples reaching that leaf

29. How are predicted class probabilities commonly derived at a tree leaf?

- A. From the class proportions of training samples in that leaf
- B. From the number of features only
- C. From the tree's plotting colors
- D. From the order of test samples

Answer: A. From the class proportions of training samples in that leaf

30. What is pruning in the context of decision trees?

- A. Reducing tree complexity by removing or limiting branches
- B. Adding a new feature to every node
- C. Scaling every feature to zero
- D. Converting the tree into a neural network

Answer: A. Reducing tree complexity by removing or limiting branches

31. What does `ccp_alpha` relate to in scikit -learn decision trees?

- A. Cost -complexity pruning
- B. Number of nearest neighbors
- C. Naive Bayes smoothing
- D. Logistic regression regularization C

Answer: A. Cost -complexity pruning

32. Why might cross -validation be used when choosing `max_depth`?

- A. To compare generalization performance across candidate depths
- B. To determine the number of class names
- C. To avoid fitting a model
- D. To turn a test set into training data

Answer: A. To compare generalization performance across candidate depths

33. What is a drawback of a single decision tree?

- A. Small changes in training data can sometimes produce a different tree
- B. It cannot predict categories
- C. It requires all features to be images
- D. It always takes hours to make one prediction

Answer: A. Small changes in training data can sometimes produce a different tree

34. What does `random_state=42` help with when a tree algorithm uses randomness?

- A. Reproducibility of results across runs
- B. Guaranteeing perfect performance
- C. Increasing the number of labels

D. Replacing cross -validation

Answer: A. Reproducibility of results across runs

35. What should you examine if a tree predicts almost every sample as the majority class?

A. Class imbalance and per -class evaluation metrics

B. Only the Python variable name

C. The screen resolution

D. Only the number of plotted nodes

Answer: A. Class imbalance and per -class evaluation metrics

36. What does `class_weight="balanced"` aim to adjust?

A. The influence of classes based on their frequencies

B. The physical weight of the computer

C. The number of input columns

D. The depth of every branch to the same value

Answer: A. The influence of classes based on their frequencies

37. Can a decision tree represent nonlinear relationships between inputs and output?

A. Yes, through combinations of feature splits

B. No, every tree is a straight line

C. Only when K equals 1

D. Only when no labels are provided

Answer: A. Yes, through combinations of feature splits

38. Why may a tree with depth 1 be too simple for a complex task?

A. It makes only one split before predicting

B. It has too many leaves

C. It memorizes every training example

D. It contains multiple forests

Answer: A. It makes only one split before predicting

39. What is a decision stump?

A. A decision tree with a single split

B. A tree with no target variable

C. A random forest with one hundred trees

D. A tree used only for regression

Answer: A. A decision tree with a single split

40. What does `model.get_depth()` report?

A. The depth of the fitted tree

B. The number of training features

C. Test accuracy

D. The number of classes

Answer: A. The depth of the fitted tree

41. What does `model.get_n_leaves()` report?

A. The number of leaf nodes

B. The number of training samples

- C. The number of feature columns
- D. The number of incorrect predictions

Answer: A. The number of leaf nodes

42. Why should you keep a separate test set when tuning a decision tree?

- A. To obtain a final evaluation on data not used for tuning
- B. To increase training accuracy automatically
- C. To ensure the tree has only two leaves
- D. To avoid calculating predictions

Answer: A. To obtain a final evaluation on data not used for tuning

43. Which evaluation metric is appropriate for a decision tree regressor?

- A. Mean absolute error
- B. Classification accuracy only
- C. Precision for a positive class
- D. Confusion matrix only

Answer: A. Mean absolute error

44. Which evaluation metric is appropriate for a decision tree classifier?

- A. F1 -score
- B. Mean absolute error only
- C. Mean squared error only
- D. R -squared only

Answer: A. F1 -score

45. What is data leakage when preparing a tree model?

- A. Information unavailable at prediction time enters training or evaluation
- B. Data is stored in two CSV files
- C. The tree has many leaf nodes
- D. The model uses an integer feature

Answer: A. Information unavailable at prediction time enters training or evaluation

46. A hospital model uses "discharge diagnosis" to predict disease at admission. What is the concern?

- A. The feature contains information that would not yet be available at admission
- B. The feature is categorical
- C. The tree must have three levels
- D. The target must be continuous

Answer: A. The feature contains information that would not yet be available at admission

47. Which parameter directly restricts the maximum number of leaf nodes?

- A. max_leaf_nodes
- B. n_neighbors
- C. alpha in MultinomialNB
- D. n_estimators

Answer: A. max_leaf_nodes

48. What does an axis -aligned split test in a standard scikit -learn decision tree?

- A. One feature against a threshold at a time

- B. All features against a single combined coefficient
- C. The target against the test accuracy
- D. Only the first training sample

Answer: A. One feature against a threshold at a time

49. Why might one -hot encoding be used before fitting a tree with categorical input columns in scikit - learn?

- A. Standard scikit -learn tree estimators require numeric feature arrays
- B. Trees require every feature to have mean zero
- C. One -hot encoding creates target labels automatically
- D. It guarantees the smallest possible tree

Answer: A. Standard scikit -learn tree estimators require numeric feature arrays

50. When is a small decision tree a useful choice?

- A. When clear, inspectable rules are valuable
- B. When no labeled examples are available
- C. When the model must ignore all features
- D. When the only goal is to store raw files

Answer: A. When clear, inspectable rules are valuable

4. Random Forest

1. What is a random forest classifier?

- A. An ensemble of decision trees used together for classification
- B. A single logistic equation
- C. A nearest -neighbor search method
- D. A text tokenizer

Answer: A. An ensemble of decision trees used together for classification

2. Why are multiple trees combined in a random forest?

- A. To make predictions more stable than a single tree in many settings
- B. To eliminate the need for data
- C. To ensure every tree is identical
- D. To convert classification into clustering

Answer: A. To make predictions more stable than a single tree in many settings

3. Which scikit -learn class is used for random forest classification?

- A. RandomForestClassifier
- B. DecisionTreeRegressor
- C. KNeighborsClassifier
- D. GaussianNB

Answer: A. RandomForestClassifier

4. Which class is used to predict continuous values with a random forest?

- A. RandomForestRegressor
- B. RandomForestClassifier
- C. LogisticRegression
- D. BernoulliNB

Answer: A. RandomForestRegressor

5. Which import statement is appropriate for random forest classification?

- A. `from sklearn.ensemble import RandomForestClassifier`
- B. `from sklearn.tree import RandomForestClassifier`
- C. `from sklearn.naive_bayes import RandomForestClassifier`
- D. `from sklearn.neighbors import RandomForestClassifier`

Answer: A. `from sklearn.ensemble import RandomForestClassifier`

6. What does `n_estimators=100` specify?

- A. Build 100 trees
- B. Use 100 input features
- C. Create 100 target classes
- D. Train for 100 epochs

Answer: A. Build 100 trees

7. How does a random forest commonly produce a classification prediction?

- A. By combining the predictions of its trees
- B. By using only the deepest tree
- C. By finding the nearest training sample
- D. By averaging the input feature values

Answer: A. By combining the predictions of its trees

8. How does a random forest regressor commonly combine tree outputs?

- A. It averages their numerical predictions
- B. It chooses the alphabetically first prediction
- C. It multiplies all predicted values
- D. It reports only the highest tree value

Answer: A. It averages their numerical predictions

9. What is bootstrap sampling in a random forest?

- A. Sampling training rows with replacement for individual trees
- B. Deleting all repeated feature values
- C. Scaling features to a common range
- D. Sampling only from the test set

Answer: A. Sampling training rows with replacement for individual trees

10. What does "with replacement" mean?

- A. The same training row can be selected more than once in a sample
- B. Every selected row is permanently removed
- C. Test rows replace training rows
- D. Missing values are replaced by zero

Answer: A. The same training row can be selected more than once in a sample

11. Why does a random forest consider random subsets of features at splits?

- A. To introduce diversity among trees
- B. To make all trees identical
- C. To avoid using target labels
- D. To force one feature to determine every split

Answer: A. To introduce diversity among trees

12. What is a key distinction between a single decision tree and a random forest?

- A. A forest combines multiple trees
- B. A forest never uses feature splits
- C. A single tree requires no labeled data
- D. A forest can only perform regression

Answer: A. A forest combines multiple trees

13. What does `model.fit(X_train, y_train)` do?

- A. Fits the forest using labeled training examples
- B. Predicts test labels without training
- C. Generates a confusion matrix only
- D. Removes the target column automatically

Answer: A. Fits the forest using labeled training examples

14. What does `model.predict(X_test)` return from a forest classifier?

- A. Predicted class labels
- B. Only the number of trees
- C. Training indices
- D. The names of all leaf nodes

Answer: A. Predicted class labels

15. What does `model.predict_proba(X_test)` provide?

- A. Estimated probabilities for each class
- B. Only a list of feature names
- C. Mean squared error
- D. The bootstrap samples themselves

Answer: A. Estimated probabilities for each class

16. What does `random_state=42` help make reproducible?

- A. Random choices in fitting and their resulting model
- B. The real -world data -generating process
- C. Perfect test accuracy
- D. The number of input features

Answer: A. Random choices in fitting and their resulting model

17. Why is feature scaling usually not essential for a standard random forest?

- A. Tree splits rely on thresholds rather than distances between full samples
- B. The forest scales all features automatically to mean zero
- C. Forests use only categorical features
- D. Forests cannot accept scaled data

Answer: A. Tree splits rely on thresholds rather than distances between full samples

18. What is an out-of-bag (OOB) sample for a particular tree?

- A. A training sample not selected for that tree's bootstrap sample
- B. A sample from the final test set
- C. A sample with a missing label

D. A sample with only one feature

Answer: A. A training sample not selected for that tree's bootstrap sample

19. What does `oob_score=True` enable when bootstrap sampling is used?

A. An internal performance estimate using out -of-bag training samples

B. Automatic removal of all outliers

C. A guarantee of unbiased real -world performance

D. A switch to logistic regression

Answer: A. An internal performance estimate using out -of-bag training samples

20. Should OOB performance automatically replace a truly independent final test evaluation?

A. No

B. Yes, in all cases

C. Yes, but only for regression

D. Yes, if there are more than ten trees

Answer: A. No

21. What does `max_depth` control in a random forest?

A. The maximum depth of its individual trees

B. The number of forests

C. The number of classes

D. The size of the test set

Answer: A. The maximum depth of its individual trees

22. What can increasing `min_samples_leaf` do?

A. Make individual trees less focused on tiny groups of training samples

B. Force every tree to have more depth

C. Set the number of trees to one

D. Remove all continuous features

Answer: A. Make individual trees less focused on tiny groups of training samples

23. What does `max_features` influence?

A. How many features are considered when looking for a split

B. How many target labels can be predicted

C. How many predictions can be made

D. How many CSV files can be loaded

Answer: A. How many features are considered when looking for a split

24. What is a likely result if all trees are nearly identical?

A. Combining them offers less benefit from diversity

B. Their votes become invalid Python syntax

C. The number of classes doubles

D. The forest becomes an unsupervised model

Answer: A. Combining them offers less benefit from diversity

25. Which task is suitable for `RandomForestClassifier`?

A. Predicting whether a transaction is fraudulent

B. Predicting an exact continuous sales amount

- C. Compressing an image file
- D. Sorting patient IDs

Answer: A. Predicting whether a transaction is fraudulent

26. Which task is suitable for RandomForestRegressor?

- A. Predicting a patient's length of stay in days as a numerical value
- B. Assigning a document to a topic category
- C. Classifying a transaction as fraudulent
- D. Identifying an email as spam

Answer: A. Predicting a patient's length of stay in days as a numerical value

27. What is one reason a random forest may outperform a single deep tree on unseen data?

- A. Combining varied trees can reduce sensitivity to individual training examples
- B. Every tree is guaranteed to be perfectly accurate
- C. It removes the need for a training set
- D. It uses the test labels during fitting

Answer: A. Combining varied trees can reduce sensitivity to individual training examples

28. What is a disadvantage of a forest compared with a small single tree?

- A. Its overall prediction process is harder to explain as a short set of rules
- B. It cannot use numeric features
- C. It cannot make classification predictions
- D. It has no adjustable parameters

Answer: A. Its overall prediction process is harder to explain as a short set of rules

29. What does feature_importances_ report in a standard fitted scikit -learn forest?

- A. Impurity -based feature importance scores
- B. Proof of cause and effect
- C. Predictions for the test set
- D. The number of missing values per feature

Answer: A. Impurity -based feature importance scores

30. Why can impurity -based feature importance be misleading?

- A. It can favor certain feature types and does not establish causation
- B. It is always identical for every feature
- C. It is calculated from test labels only
- D. It cannot be computed for forests

Answer: A. It can favor certain feature types and does not establish causation

31. What is permutation importance based on?

- A. Measuring performance changes after shuffling a feature's values
- B. Counting the letters in feature names
- C. Deleting the target labels
- D. Replacing all trees with one tree

Answer: A. Measuring performance changes after shuffling a feature's values

32. Why should feature importance ideally be examined on suitable held -out data?

- A. To see whether the feature helps predictions beyond the training examples

- B. To force all importance values to be equal
- C. To avoid making any predictions
- D. To increase the number of classes

Answer: A. To see whether the feature helps predictions beyond the training examples

33. What does `class_weight="balanced"` address in a forest classifier?

- A. Differences in class frequency
- B. Different physical units of features
- C. Different file sizes of datasets
- D. Different depths of plotted trees only

Answer: A. Differences in class frequency

34. A fraud dataset contains 99% legitimate transactions. Which evaluation is especially useful?

- A. Precision and recall for fraud
- B. Training accuracy alone
- C. Number of trees alone
- D. Feature count alone

Answer: A. Precision and recall for fraud

35. Why can overall accuracy be misleading in that fraud dataset?

- A. A model can predict "legitimate" almost always and still achieve high accuracy
- B. Accuracy cannot be calculated for forests
- C. Fraud must be a continuous target
- D. Forests ignore class imbalance by definition

Answer: A. A model can predict "legitimate" almost always and still achieve high accuracy

36. What may happen as `n_estimators` increases substantially?

- A. Fitting and prediction may require more computation
- B. The target becomes continuous
- C. The dataset gains new observations automatically
- D. Every tree becomes shallower by definition

Answer: A. Fitting and prediction may require more computation

37. Does adding more trees always guarantee better test performance?

- A. No
- B. Yes
- C. Yes, but only when there are two features
- D. Yes, if the training accuracy is high

Answer: A. No

38. What does `n_jobs=-1` commonly request in a scikit-learn forest?

- A. Use all available CPU workers for supported parallel computation
- B. Train one less tree than requested
- C. Remove the last feature
- D. Use GPU acceleration automatically

Answer: A. Use all available CPU workers for supported parallel computation

39. Why should you use cross -validation when selecting forest hyperparameters?

- A. To compare candidate settings on multiple training/validation splits
- B. To obtain the true labels for future samples
- C. To guarantee a specific test score
- D. To eliminate the need for training

Answer: A. To compare candidate settings on multiple training/validation splits

40. Which code correctly constructs and fits a forest classifier?

- A. `model = RandomForestClassifier(n_estimators=100); model.fit(X_train, y_train)`
- B. `model = RandomForestClassifier.fit(X_test)`
- C. `model = RandomForestClassifier(y_train); model.predict(X_train, y_train)`
- D. `model = fit(RandomForestClassifier, X_test)`

Answer: A. `model = RandomForestClassifier(n_estimators=100); model.fit(X_train, y_train)`

41. What does `train_test_split(..., stratify=y)` help preserve in classification?

- A. Approximate class proportions across splits
- B. Feature scaling parameters
- C. Identical rows in both splits
- D. The same prediction for every class

Answer: A. Approximate class proportions across splits

42. What is data leakage in a forest -based project?

- A. Training or model selection uses information that would not be available at prediction time
- B. Using more than one tree
- C. Using a categorical target
- D. Setting `random_state`

Answer: A. Training or model selection uses information that would not be available at prediction time

43. A hospital dataset includes an outcome recorded after treatment, but the model is meant to predict at admission. What should you do?

- A. Exclude that future -information feature from admission -time prediction
- B. Use it because it improves training accuracy
- C. Duplicate it across several columns
- D. Increase `n_estimators` until it disappears

Answer: A. Exclude that future -information feature from admission -time prediction

44. Can a random forest model interactions between features?

- A. Yes, tree paths can combine splits on multiple features
- B. No, every feature is considered completely independently
- C. Only if all features are standardized
- D. Only when there are exactly two classes

Answer: A. Yes, tree paths can combine splits on multiple features

45. How does a forest typically behave when asked to predict a regression value far outside the range of observed target values?

- A. It generally has limited ability to extrapolate beyond observed target ranges
- B. It always extrapolates perfectly

- C. It changes into a linear model
- D. It refuses to predict any value

Answer: A. It generally has limited ability to extrapolate beyond observed target ranges

46. Why might one-hot encoding be used for categorical input features with scikit-learn random forests?

- A. The estimator expects numeric feature input
- B. It forces every tree to have exactly one leaf
- C. It guarantees feature importance is causal
- D. It creates the target variable

Answer: A. The estimator expects numeric feature input

47. What is a reasonable way to compare a decision tree and a random forest?

- A. Evaluate both on the same appropriate validation or test data and metrics
- B. Compare only the length of their Python class names
- C. Compare training accuracy only
- D. Assume the forest always wins

Answer: A. Evaluate both on the same appropriate validation or test data and metrics

48. What does a random forest's predicted probability of 0.8 require before use as a reliable risk estimate?

- A. Evaluation and possibly calibration on representative data
- B. More trees alone, with no evaluation
- C. Converting every feature to a string
- D. A deeper single tree

Answer: A. Evaluation and possibly calibration on representative data

49. What is the forest's use of randomness intended to help create?

- A. Diversity among its trees
- B. Random target labels for test samples
- C. Missing values in every feature
- D. An identical tree in every position

Answer: A. Diversity among its trees

50. When is a random forest a practical baseline?

- A. For labeled tabular data with potentially nonlinear relationships
- B. When the task has no inputs and no target
- C. When the only goal is to edit images
- D. When a single fixed linear coefficient is required for every feature

Answer: A. For labeled tabular data with potentially nonlinear relationships

5. Logistic Regression

1. Despite its name, what is logistic regression commonly used for?

- A. Classification
- B. Image compression
- C. Clustering only
- D. Sorting rows

Answer: A. Classification

2. In binary logistic regression, what is commonly predicted before applying a decision threshold?

- A. A probability -like estimate for a class
- B. The nearest training sample
- C. The depth of a tree
- D. The number of clusters

Answer: A. A probability -like estimate for a class

3. Which scikit -learn class implements logistic regression?

- A. LogisticRegression
- B. LinearRegression
- C. LogisticClassifierTree
- D. GaussianNB

Answer: A. LogisticRegression

4. Which import statement is appropriate?

- A. from sklearn.linear_model import LogisticRegression
- B. from sklearn.tree import LogisticRegression
- C. from sklearn.neighbors import LogisticRegression
- D. from sklearn.metrics import LogisticRegression

Answer: A. from sklearn.linear_model import LogisticRegression

5. Which is a suitable binary logistic regression target?

- A. Disease present or absent
- B. Exact temperature in degrees Celsius
- C. Exact product price
- D. Exact patient weight

Answer: A. Disease present or absent

6. What does model.fit(X_train, y_train) do?

- A. Estimates model parameters using labeled training data
- B. Returns test accuracy without training
- C. Converts the model into KNN
- D. Creates a decision tree plot

Answer: A. Estimates model parameters using labeled training data

7. What does model.predict(X_test) return?

- A. Predicted class labels
- B. Only coefficient values
- C. The input feature names
- D. The training data

Answer: A. Predicted class labels

8. What does model.predict_proba(X_test) return?

- A. Estimated probabilities for each class
- B. Only the confusion matrix
- C. Euclidean distances
- D. Training -set indices

Answer: A. Estimated probabilities for each class

9. What function maps a real -valued score into a value between 0 and 1 in binary logistic regression?

- A. Sigmoid function
- B. Gini function
- C. Euclidean function
- D. Bootstrap function

Answer: A. Sigmoid function

10. What is a decision threshold?

- A. A probability cutoff used to convert a score into a class prediction
- B. The number of training features
- C. The number of model coefficients
- D. The number of validation folds

Answer: A. A probability cutoff used to convert a score into a class prediction

11. With a 0.5 threshold, what class is usually assigned when the estimated positive -class probability is 0.8?

- A. Positive
- B. Negative
- C. Both classes
- D. No class

Answer: A. Positive

12. Why might a medical screening system use a threshold below 0.5?

- A. To identify more potential positive cases, while potentially increasing false alarms
- B. To remove all false positives and false negatives
- C. To make the model a random forest
- D. To eliminate the need for validation

Answer: A. To identify more potential positive cases, while potentially increasing false alarms

13. What generally happens to recall for the positive class when the positive decision threshold is lowered?

- A. It may increase
- B. It must become zero
- C. It always remains identical
- D. It becomes undefined for every dataset

Answer: A. It may increase

14. What generally happens to the number of false positives when the positive threshold is lowered?

- A. It may increase
- B. It must become zero
- C. It always remains identical
- D. It becomes equal to the number of features

Answer: A. It may increase

15. What does a positive coefficient suggest, with other features held fixed?

- A. Larger values of that feature increase the model's log -odds of the positive class
- B. Larger values always decrease predicted probability
- C. The feature is necessarily causal

D. The feature is ignored

Answer: A. Larger values of that feature increase the model's log -odds of the positive class

16. Does a larger positive coefficient always mean a feature is more important than another feature measured on a different scale?

A. No, feature units and scaling affect coefficient magnitudes

B. Yes, regardless of units

C. Yes, but only if there are three classes

D. No, because coefficients cannot be positive

Answer: A. No, feature units and scaling affect coefficient magnitudes

17. Why is feature scaling often helpful for logistic regression?

A. It can improve optimization and make regularization act more comparably across features

B. It converts classification into regression

C. It guarantees perfect calibration

D. It changes the number of labels

Answer: A. It can improve optimization and make regularization act more comparably across features

18. Which preprocessing tool is commonly used to standardize numeric features?

A. StandardScaler

B. CountVectorizer for all numeric data

C. plot_tree

D. confusion_matrix

Answer: A. StandardScaler

19. Which code structure combines scaling and logistic regression appropriately?

A. Pipeline([("scaler", StandardScaler()), ("model", LogisticRegression())])

B. Pipeline([("model", LogisticRegression()), ("scaler", StandardScaler())])

C. LogisticRegression(StandardScaler())

D. StandardScaler.fit(LogisticRegression())

Answer: A. Pipeline([("scaler", StandardScaler()), ("model", LogisticRegression())])

20. Why should preprocessing be fitted within training folds during cross -validation?

A. To prevent validation information leaking into training

B. To ensure the validation set has no rows

C. To increase the number of classes

D. To make all coefficients equal

Answer: A. To prevent validation information leaking into training

21. What does regularization generally do?

A. Discourages excessively large coefficients

B. Adds more target classes

C. Selects the nearest neighbors

D. Creates deeper tree branches

Answer: A. Discourages excessively large coefficients

22. In scikit -learn logistic regression, what does a smaller C generally mean?

A. Stronger regularization

- B. Weaker regularization
- C. More trees
- D. More nearest neighbors

Answer: A. Stronger regularization

23. What does a larger C generally mean?

- A. Weaker regularization
- B. Stronger regularization
- C. Fewer training rows
- D. More classes by definition

Answer: A. Weaker regularization

24. What can L1 regularization encourage?

- A. Some coefficients becoming exactly zero
- B. More trees in an ensemble
- C. Every feature becoming identical
- D. All probabilities becoming 0.5

Answer: A. Some coefficients becoming exactly zero

25. What does L2 regularization typically encourage?

- A. Smaller coefficients without generally forcing them all to zero
- B. A tree with two leaves
- C. Equal class frequencies
- D. A fixed number of neighbors

Answer: A. Smaller coefficients without generally forcing them all to zero

26. What is the decision boundary of basic binary logistic regression in the original feature space?

- A. Linear in the supplied features
- B. Always circular
- C. A random collection of tree branches
- D. Based only on the nearest sample

Answer: A. Linear in the supplied features

27. Can logistic regression represent a nonlinear pattern if suitable transformed features are supplied?

- A. Yes
- B. No, under any circumstances
- C. Only with bootstrap sampling
- D. Only if labels are missing

Answer: A. Yes

28. Which transformation could help model a curved relationship?

- A. Adding polynomial or interaction features
- B. Removing every input feature
- C. Replacing the target with row numbers
- D. Increasing the test set to 100%

Answer: A. Adding polynomial or interaction features

29. What are log -odds in logistic regression?

- A. The logarithm of the odds of the positive class
- B. The logarithm of the number of features
- C. The number of false positives
- D. The distance between training samples

Answer: A. The logarithm of the odds of the positive class

30. If a coefficient is zero, what does that mean within the fitted linear score?

- A. That feature makes no direct contribution to the score through that coefficient
- B. The feature must be missing from the original dataset
- C. The target is always zero
- D. The model cannot predict probabilities

Answer: A. That feature makes no direct contribution to the score through that coefficient

31. Which metric measures how many predicted positives are actually positive?

- A. Precision
- B. Recall
- C. Mean squared error
- D. R -squared

Answer: A. Precision

32. Which metric measures how many actual positives are identified?

- A. Recall
- B. Precision
- C. Number of coefficients
- D. Training sample count

Answer: A. Recall

33. Why may accuracy be misleading for rare -event classification?

- A. Predicting the majority class can give high accuracy while missing rare events
- B. Accuracy cannot be used for logistic regression
- C. Accuracy always exceeds 1
- D. It measures tree depth

Answer: A. Predicting the majority class can give high accuracy while missing rare events

34. What does a confusion matrix summarize?

- A. Counts of correct and incorrect predictions by actual and predicted class
- B. The values of model coefficients only
- C. The feature scaling means only
- D. The number of optimization iterations only

Answer: A. Counts of correct and incorrect predictions by actual and predicted class

35. What does ROC AUC evaluate?

- A. How well model scores rank positive examples above negative examples across thresholds
- B. Only performance at exactly one fixed threshold
- C. Mean absolute error
- D. The number of input features

Answer: A. How well model scores rank positive examples above negative examples across thresholds

36. What can a precision -recall curve be especially informative for?

- A. Evaluating positive -class detection in an imbalanced dataset
- B. Selecting the number of trees
- C. Measuring tree depth
- D. Computing Euclidean distance

Answer: A. Evaluating positive -class detection in an imbalanced dataset

37. Why should predicted probabilities be checked for calibration?

- A. A predicted risk of 0.8 should correspond reasonably to observed outcomes near that level
- B. Calibration changes the target from categorical to continuous
- C. Calibration guarantees zero classification errors
- D. Calibration is the same as feature scaling

Answer: A. A predicted risk of 0.8 should correspond reasonably to observed outcomes near that level

38. Which scikit -learn attribute contains fitted coefficients?

- A. `coef_`
- B. `feature_importances_`
- C. `n_neighbors_`
- D. `tree_`

Answer: A. `coef_`

39. Which attribute contains the fitted intercept?

- A. `intercept_`
- B. `max_depth_`
- C. `bootstrap_`
- D. `classes_count_`

Answer: A. `intercept_`

40. What does `model.classes_` help you identify?

- A. Which class corresponds to each column of `predict_proba`
- B. The order of training rows
- C. The names of numerical features
- D. The number of validation folds

Answer: A. Which class corresponds to each column of `predict_proba`

41. Can scikit -learn logistic regression handle multiclass classification?

- A. Yes
- B. No, only two classes are possible
- C. Only when K is odd
- D. Only when features are text strings

Answer: A. Yes

42. Which is an example of multiclass classification?

- A. Assigning a document to sports, politics, or science
- B. Predicting exact monthly revenue
- C. Predicting exact blood glucose level

D. Estimating the number of trees

Answer: A. Assigning a document to sports, politics, or science

43. What is the purpose of `class_weight="balanced"`?

- A. To weight classes inversely according to their frequency
- B. To scale each input feature to the same range
- C. To make every coefficient positive
- D. To set the decision threshold to exactly 0.8

Answer: A. To weight classes inversely according to their frequency

44. Why should you avoid fitting a logistic regression model using a feature recorded after the outcome?

- A. It may leak information unavailable at prediction time
- B. Logistic regression accepts only earlier dates as numbers
- C. The number of classes will change
- D. It creates too many trees

Answer: A. It may leak information unavailable at prediction time

45. What does convergence mean during model fitting?

- A. The optimization procedure has reached its stopping condition for parameter estimation
- B. All predictions are correct
- C. Every coefficient is zero
- D. The dataset has become balanced

Answer: A. The optimization procedure has reached its stopping condition for parameter estimation

46. If `scikit-learn` warns that logistic regression failed to converge, what is a sensible first check?

- A. Feature scaling, solver settings, and the iteration limit
- B. Whether the variable name contains "logistic"
- C. Whether `n_estimators` is at least 100
- D. Whether `K` is an odd number

Answer: A. Feature scaling, solver settings, and the iteration limit

47. What does `max_iter` control?

- A. The maximum number of optimization iterations
- B. The number of nearest neighbors
- C. The maximum tree depth
- D. The number of training features

Answer: A. The maximum number of optimization iterations

48. Why is a separate test set useful after tuning `C` and other choices?

- A. It provides a final assessment on examples not used to choose the model
- B. It increases the number of model coefficients
- C. It removes the need for training labels
- D. It guarantees perfect deployment performance

Answer: A. It provides a final assessment on examples not used to choose the model

49. When is logistic regression a sensible baseline?

- A. When you want a relatively simple classifier with inspectable coefficients and predicted probabilities
- B. When no labeled data are available

- C. When the output must be a decision tree diagram
- D. When the task is to find nearest images without labels

Answer: A. When you want a relatively simple classifier with inspectable coefficients and predicted probabilities

50. A model predicts a patient's positive -class probability as 0.72. What should determine whether this produces a positive classification?

- A. The chosen decision threshold
- B. The patient's row number
- C. The number of trees in the dataset
- D. The alphabetical order of features

Answer: A. The chosen decision threshold